



Performance evaluation and reconfiguration of supply chains based on discrete event systems formalisms

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Abstract

This paper addresses the time performance evaluation of supply chains, modeled by a class of uncertain timed discrete event systems, known as timed automata with guards. The contributions of this work introduce a novel approach to performance analysis that evaluates upper and lower bounds on completion times using advanced algorithms based on dioid algebra. Furthermore, we propose innovative methodologies for reconfiguring supply chains, integrating time-sensitive features into perturbation processes. This ensures the selection of optimal reconfiguration scenarios that enhance resilience while minimizing execution times. Compared to existing approaches, our framework prioritizes faster and more reliable task execution. A comprehensive case study illustrates the practical applicability and advantages of the proposed methods.

Keywords Supply chains · Timed discrete event systems · Timed automata with guards · Performance evaluation · Resilience

1 Introduction

Successfully modeling of a Supply Chain (SC) is a critical step in strategic planning. Clearly defining the SC is essential for companies as it helps establish their market position and chart their future course. However, the complexity and uncertainties inherent in these processes present significant challenges in their design and management (Croom et al. 2000). Enterprises must meet the demands of customers who

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increasingly expect faster delivery times, high product quality, competitive prices and ensure industrial performances (Tayur et al. 1998).

Numerous studies on logistics networks have therefore been carried out to model (Badole 2012) and guarantee performance analysis (Melo et al. 2009; Borges et al. 2016; Barzizza et al. 2023), in order to identify the most effective method for getting the products to clients.

In what concern modeling process, some authors used Petri nets (PN), as it is a powerful tool for modeling Discrete Event Systems (DES) (Wang 1998). Given that SC can also be represented by DES at high level of abstraction, it is feasible to develop a Petri net-based tool for representing SC. In (Viswanadham and Raghavan 2000), authors considered Generalized Stochastic Petri Nets (GSPN) to model SC. When talking about batch Deterministic and Stochastic Petri Nets (DSPN) models, we refer to the research conducted by Chen et al., (2005). Additionally, SC are modeled by hierarchical PN (Blackhurst et al 2008). Zegordi and Davarzani (2012) modeled SC by colored PN.

However, all the references mentioned above do not incorporate the time factor into their modeling. It is worth mentioning the study of (Elmahi et al. 2002), where SC are modeled by TEG. Other studies, such as the one presented in (Liu et al. 2007) have used Time Colored PN to model SC. They introduced seven basic patterns to capture common modeling concepts, demonstrating their combination to construct comprehensive PN. In the same context, Mahjoub et al. (2021) have dealt with SC management using Timed Colored PN.

Nevertheless, the aforementioned methodologies concentrate mostly on SC when tasks and operation durations are precisely defined. In the real world, this presumption is not always met. In order to figure out the drawbacks of existing techniques, studies such as (Ait-Oumeziane et al. 2020b) examines the uncertainty surrounding task execution. They created new formal approaches for modeling and controlling uncertain timed DES using Timed Automata with Guards (TAG). Techniques introduced in (Tamssaouet and Amari 2018) also used TAG to represent the treated SC.

After establishing a detailed model of the system using techniques such as Petri nets and timed event graphs, the next crucial step is to evaluate the system performance. While modeling provides a structured representation of the system processes and interactions, performance analysis allows us to assess how efficiently the system operates under various conditions. This involves measuring key performance metrics such as throughput, response time, and resource utilization. By leveraging tools like simulation and optimization methods, we can gain valuable insights into potential bottlenecks and system inefficiencies, ultimately ensuring that the model reflects real-world behavior and can inform practical decision-making. Citing the techniques in (Viswanadham and Raghavan 2000), authors here investigated dynamic modeling techniques for analyzing Supply Chain Networks (SCN), accounting for logistics and interface processes. Researchers like Chen et al. (2005), developed approaches for structural analysis by expanding existing models for DSPN. The work of (Blackhurst et al 2008) presents a methodology to proactively detect conflicts in SC to prevent detrimental effects on system performance. By linking hierarchical levels and identifying conflicts, it offers valuable insight into managing distributed entities efficiently. Elmahi et al. (2002) used

the application of (Max, +) algebra to estimate performances by describing state equations. The determination of production optimal dates is proposed in this work and the control of the system is done by the supervision of these dates. The studies in (Mahjoub et al. 2021) have dealt with (Max, +) equations. The approach also evaluates delivery times and optimizes vehicle deployment to ensure efficient Supply Chain management and validated through various scenarios. Furthermore, a wide range of timed DES can be modeled using (Max, +) automata (Boukra et al. 2012), which operate within the (Max, +) algebra. The research is based on the use of recursive equations in both (Max, +) and (Min, +) algebras to effectively describe extremal behaviors and simplify the evaluation of performance metrics like maximum and minimum execution times. The work of Mahmoud et al. (2024) focused mainly on conceptual models for integrating the circular economy into SC management. It mainly addressed environmental factors linked to the challenges of climate change and waste management, without proposing formal solutions for anticipating and managing operational risks. Ait-Oumeziane et al. (2020b) proposed alternative representations employing recursive equations in (Max, +) and (Min, +) algebras, delineating both worst and the optimistic behaviors of SC. Along the same lines, Ait-Oumeziane et al. (2020a) shed light on evaluating time performance of Distributed Control Systems (DCS) using dioid algebra, focusing on estimating response time and reactivity. Also, the estimation of the upper bound of response time for client/server networked automation architectures by using (Max, +) algebra is discussed in (Tamssaouet and Amari 2018). Nevertheless, this methodology considers simple examples of networked control systems with independent Programmable Logic Controllers (PLCs), which overlooked several important delays in the actual response time. Recent researches (Muhammad Shamrooz et al. 2024) have introduced a fuzzy control strategy for managing nonlinear supply chain systems, focusing on the impact of lead times. The authors have used maximal overlapping rules groups (MRRG) for stability and reducing variable fluctuations with membership functions. The approach aims to achieve stability and smooth transitions in the SC. Although this method offers a robust solution for stochastic delays and demands, it can be complex to implement in highly dynamic systems or with even more complex uncertainties.

While these studies effectively address the issue of modeling and performance analysis, they are still restricted when it comes to examine the resilience and reconfiguration of Supply Chains in response to disruptions and unexpected interruptions.

Actually, resilience through reconfiguration involves adapting and reorganizing the system to maintain optimal performance in the long term after disruptions, without requiring as many resources or initial complexity. Numerous studies have examined methods and strategies for designing and reconfiguring global supply chains to adapt to changing market dynamics. For instance, Li et al. (2023) use analytical approaches to assess and improve resilience. It relies on statistical and bibliometric analyses to identify trend provided solutions that allow supply chains to adapt to disruptions or anomalies, especially in maritime transportation. Meixell and Gargeya (2005) highlighted key approaches to global supply chain reconfiguration, emphasizing responsiveness to market shifts. Similarly, Chen et al. (2007) employed scenario planning to address reconfiguration in emergency logistics, specifically during flood events. Zhu et al. (Zhu et al. 2005) expanded on this by investigating supply chain

reconfiguration aimed at achieving sustainability objectives, with a particular focus on minimizing environmental impact. Additionally, the research of Schuh and Lunze (2016) provided insight into the process of developing an active fault-tolerant control strategy for DES represented by deterministic input/output automata and measurable states. They incorporate diagnostic and reconfiguration units to detect faults and adjust the controller. Niguez et al. (2017) have created a time-constrained model of the system behavior in order to derive a reconfigured control law based on the quickest task execution time after diagnosis.

Nonetheless, the referenced studies did not explicitly discuss the choice of an optimal reconfiguration. For this, the current research is dedicated to overcome these constraints, by identifying appropriate transitions between configurations, while also ensuring the selection of an optimal reconfiguration. This ensures that the chosen transitions promote longer-term resilience, maximizing the efficiency and robustness of the SC.

The contribution presented in this paper expands upon the results provided in both works (Niguez et al. 2017) and (Ait-Oumeziane et al. 2020a). It begins by modeling SC using uncertain class of DES, known as timed automata with guards. Actually, TAGs are specifically designed to model temporal behavior and event-driven changes. They capture time-dependent processes and enable the precise representation of deadlines, delays, or waiting times that are essential for the operation of supply chains. For example, a timed automaton (Subbiah et al. 2013) can model the time-dependent state transitions of inventory levels, production cycles, or transportation logistics. The combination of timed automata and guards allows for a more accurate and detailed modeling of supply chain performance, accounting for both time constraints and delays, which are often not sufficiently captured by traditional optimization techniques. The methodology then evaluates the temporal performance of the system and estimates the maximum and minimum bounds of product routing using properties of dioid algebra. Indeed, performance analysis plays a crucial role in guiding the effective reconfiguration of the SC by providing the necessary data and insights. It is worth mentioning that dioid algebra allows for efficient composition and propagation of minimum-cost or maximum-flow decisions through the supply chain, even as the system size increases. It can scale to large networks, whereas traditional Linear Programming (LP) models (Preusser et al. 2005; Alawneh et al. 2014) and Mixed-Integer Programming (MIP) representations (Li et al. 2024) may become computationally intractable. In addition, it provides a robust mathematical framework for optimizing the flow of resources and information across the supply chain, ensuring that the reconfiguration of supply chain processes is efficient. It can handle operations such as minimizing delivery time or maximizing throughput in the face of event-driven changes.

Actually, the combination of timed automata with guards and dioid algebra provides a powerful and formal framework for performance evaluation in logistics chains, offering several distinct advantages compared to other methods in the literature. Specifically, our approach enables: (i) Precise, time-based modeling that captures delays and time dependencies, (ii) Formal and structured analysis leading to more reliable performance evaluations, (iii) Modularity and decomposability, enabling independent optimization of logistics components and (iv) Real-time adaptation to disruptions or

changes in the logistics system. These characteristics collectively make our approach more suitable for time-sensitive performance evaluation, optimization, and adaptability, particularly in complex and evolving supply chain scenarios. While direct quantitative comparisons with all existing techniques may vary depending on specific case studies, our approach provides a structured and scalable method that addresses key limitations of traditional models.

The motivation of this proposal is mainly based on modeling and generating permissible reconfigurations of the SC with the existence of anomalies and malfunctions during operation, with the aim of selecting an optimal reconfiguration.

The main scientific contributions of this study are: (1) Modeling Supply Chains using uncertain discrete event systems, using specifically timed automata with guards; (2) Development of a method to estimate the maximum and minimum bounds for product routing using dioid algebra properties; (3) Proposal of a reconfiguration approach to generate optimized reconfigurations in the presence of anomalies or malfunctions in Supply Chains; (4) Definition of appropriate transitions between different configurations to select the optimal reconfiguration; (5) Implementation of the theoretical results into a real-world application.

These contributions offer several advantages for Supply Chain management, providing a robust method for modeling uncertain and complex systems through timed automata with guards. This approach more accurately reflects real-world conditions where task durations and operations are not always deterministic. By incorporating dioid algebra, the model allows for a comprehensive performance analysis of SCs, estimating maximum and minimum bounds for product routing and helping identify potential bottlenecks and inefficiencies. Furthermore, the proposed reconfiguration strategy enables SCs to adapt to disruptions or malfunctions, improving resilience without requiring extensive resources or complex changes. The focus on selecting optimal reconfigurations based on performance data ensures more efficient use of resources while maintaining system robustness. Ultimately, this approach contributes to more resilient, adaptive, and efficient SC, capable of maintaining optimal performance even in the face of uncertainties and disruptions.

Figure 1 provides an overview of the key stages involved in this research, from the initial modeling of the Supply Chain to the final reconfiguration process.

This paper is organized as follows: Sect. 2 recalls some dioid formalisms and TAG concepts to model SC. In Sect. 3, the behavior and performance analysis of TAG using dioid algebra is discussed. Section 4 details the SC reconfiguration method by selecting the appropriate scenario to guarantee optimal functioning in the presence of disturbances. Section 5 incorporates a case study that illustrates the theoretical propositions.

2 Preliminaries

To facilitate understanding, some of the theoretical concepts involved in this study, including dioid theory and timed automata with guards, are represented in this section. It also serves as a reminder of the essential representations required to assess completion dates, which are crucial for evaluating system performance at a later stage.

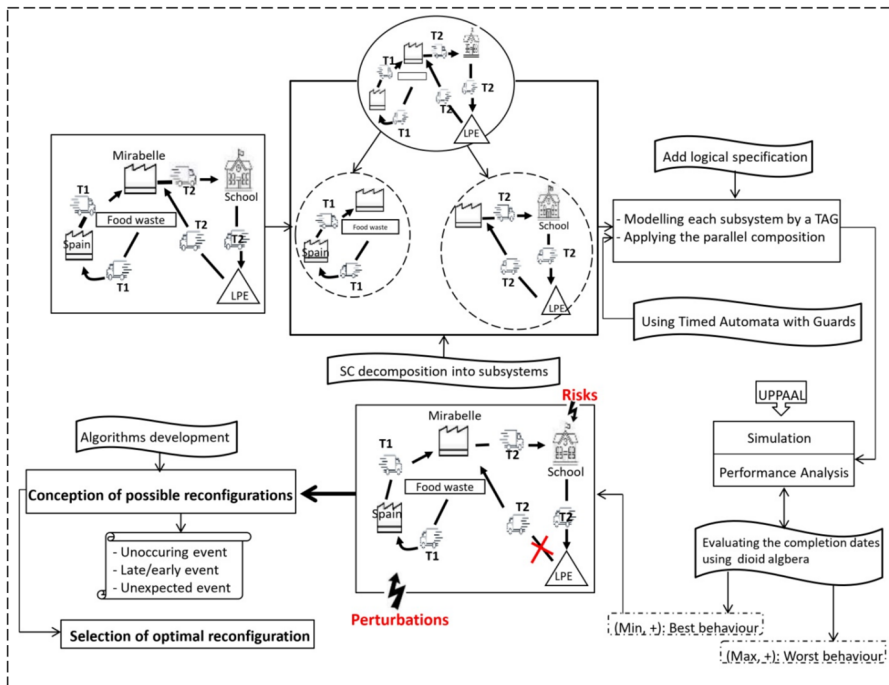


Fig. 1 Conceptual diagram of methodologies

2.1 Dioid algebra formalisms

A dioid (Baccelli et al. 1992) is defined by tuple $(D, \oplus, \otimes)$, where D is a set equipped with an associative and commutative operation $(\oplus)$ that has a neutral element denoted by ε , such that for any a in D , $a \oplus \varepsilon = \varepsilon \oplus a = a$. The second operation $(\otimes)$ of the dioid is associative, and distributive over $(\oplus)$ from both left and right. It features a neutral element e and an absorbing element ε , i.e.: $\forall a \in D, a \otimes \varepsilon = \varepsilon \otimes a = \varepsilon$. Furthermore, the operation $(\oplus)$ is idempotent, meaning that for any a in D , $a \oplus a = a$.

A dioid is called commutative if the operation $(\otimes)$ is also commutative. A dioid is considered complete if the operation $(\otimes)$ is right and left distributive over infinite sums, which means, $\forall b \in D, \forall A \subseteq D : b \otimes (\oplus_{a \in A} x) = \oplus_{a \in A} (b \otimes a)$ and $(\oplus_{a \in A} x) \otimes b = \oplus_{a \in A} (a \otimes b)$.

The $(\text{Min}, +)$ commutative dioid, commonly referred to as Min-Plus algebra, is represented by $\overline{\mathbb{R}}_{\text{min}} = (\mathbb{R} \cup \{+\infty\}, \oplus, \otimes)$. The operation $(\oplus)$ corresponds to the *min* function, i.e. $\forall a, b \in \overline{\mathbb{R}}_{\text{min}} : a \oplus b = \min(a, b)$. The second operation $(\otimes)$ is similar to the conventional addition, i.e. $\forall a, b \in \overline{\mathbb{R}}_{\text{min}} : a \otimes b = a + b$.

Some for $(\text{Max}, +)$, it is defined by $\overline{\mathbb{R}}_{\text{max}} = (\mathbb{R} \cup \{-\infty\}, \oplus, \otimes)$, such that: the operation $(\oplus)$ corresponds to the *max* function, in other words, $\forall a, b \in \overline{\mathbb{R}}_{\text{max}} : a \oplus b = \max(a, b)$. For the multiplication operation $(\otimes)$, it is equivalent to the classical addition, i.e. $\forall a, b \in \overline{\mathbb{R}}_{\text{max}} : a \otimes b = a + b$.

The set of $n \times m$ matrices over $\overline{\mathbb{R}}_{min}$ or $\overline{\mathbb{R}}_{max}$, in a matrix dioid is defined as $D^{n \times m}$, with $n, m \in \mathbb{N}$. The operations for all $A, B \in D^{n \times m}$ and $C \in D^{m \times p}$, are defined as:

$$(A \oplus B)_{ij} = A_{ij} \oplus B_{ij}, \forall i = 1, \dots, n, \forall j = 1, \dots, m$$

$$(A \otimes C)_{ij} = \oplus_{k=1}^m A_{ik} \otimes C_{kj}, \forall i = 1, \dots, n, \forall j = 1, \dots, p$$

2.2 Timed automata with guards

Timed Automata with Guards are an extension of timed automata, which are formal models used to describe the behavior of logistics networks or embedded systems subject to timing constraints. TAG allow the definition of conditional transitions based on guards, which are logical expressions over the system variables and temporal clocks. These guards determine the conditions under which a transition can occur, thereby introducing additional flexibility in modelling the timed behaviors of systems.

Definition 1 (Cassandras and Lafortune 2008). A TAG, denoted by G , is a 6-tuple $G = (S, E, S_0, S_f, Tra, C)$, such that:

- S : is the set of states;
- E : is the set of events;
- $S_0 \subset S$: is the set of initial states;
- $S_f \subset S$: is the set of final states;
- Tra : is the set of timed transitions of the automaton, where:

$$Tra \subseteq S \times \mathcal{C}(C) \times E \times 2^C \times S$$

Noting that $\mathcal{C}(C)$ is the set of admissible constraints for the clocks in the set C ;

- $C = c_1, \dots, c_n$: is the set of clocks, with $c_i(t) \in \mathbb{R}^+$.

Remarks The set Tra of timed transitions is interpreted as follows:

- If $(s_{in}, guard, e, reset, s_{out}) \in Tra$, then there is a transition from s_{in} to s_{out} with the complete label $(e, guard, reset)$, where $guard \in \mathcal{C}(C)$, $e \in E$ and $reset \subseteq C$.
- At the initialization of the system, all clocks are initialized to 0.
- “reset” corresponds to the subset of clocks that will be reset upon the firing of the transition. This mechanism facilitates the modelling of systems where the duration is specified for sequences of events.
- The constraints on the clock are temporal conditions between different clock values, guiding the system behavior.

Example 1 To enhance clarity and illustrate the aforementioned notations, we provide a graphical representation in Fig. 2. It is about a SC modelled by a TAG.

The process initiates with the controllable event E_1 , succeeded by a sequence of uncontrollable events a, b, e , and d . The system has the capability of being reset with

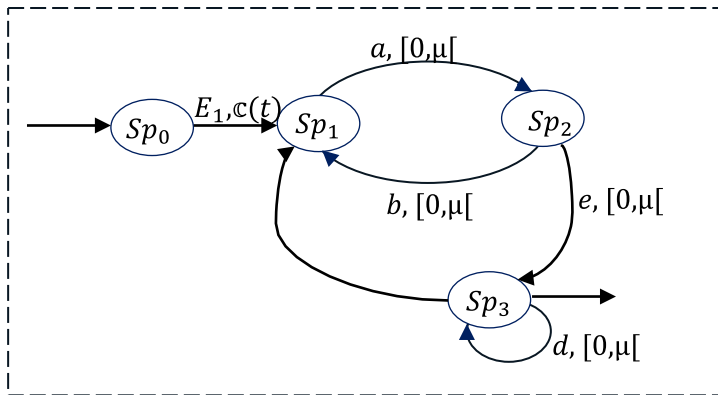


Fig. 2 Example of timed automata with guards $G(P_1)$

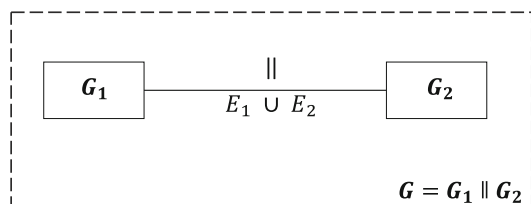
a controllable event E_2 . State Sp_0 is designated as the initial state, while state Sp_3 is marked as the final state. In this scenario, each uncontrollable event is anticipated to happen within an upper bound of μ time units, whereas each controllable event is assumed to occur immediately at the current clock value $c(t)$ upon entering a new state (with transitions intervals denoted as $[c(t), c(t)]$). However, to maintain clarity, the notation $\mathbb{c}(t)$ is utilized instead of $[c(t), c(t)]$ in the graphical representations of TAG.

2.3 Parallel composition of timed automata with guards

The parallel composition refers to the technique of allowing the modeling of systems where multiple concurrent processes interact with each other according to specified timing constraints and synchronization rules. Each component automaton represents the behavior of a subsystem, and their parallel composition captures the combined behavior of all subsystems. Figure 3 depicts the representation of an interconnection type (parallel composition) of two TAG.

Subsequently, we consider two TAG: $G_1 = (S_1, E_1, S_{0,1}, S_{f,1}, Tra_1, C_1)$ and $G_2 = (S_2, E_2, S_{0,2}, S_{f,2}, Tra_2, C_2)$. The parallel composition of G_1 and G_2 is illustrated in the Fig. 3, given the following automaton:

Fig. 3 Representation of the parallel composition



$G_{1||2} = \mathcal{Ac}(S_1 \times S_2, E_1 \times E_2, S_{0,1} \times S_{0,2}, S_{m,1} \times S_{m,2}, Tra_{1||2}, C_1 \cup C_2)$, where $\mathcal{Ac}$ refers to accessible states and $Tra_{1||2}$ is described as below:

$$Tra_{1||2} \subseteq (S_1 \times S_2) \times \mathcal{C}(C)_{1||2} \times E_1 \cup E_2 \times 2^{C_1 \cup C_2} \times (S_1 \times S_2)$$

- For all $e \in E_1 \cap E_2$, if:
 $((s_{in,i}, guard_i, e, reset_i, s_{out,i}) \in Tra_i \text{ for } i = 1, 2, \text{ then}$

$$((s_{in,1}, s_{in,2}), guard_1 \wedge guard_2, e, reset_1 \cup reset_2, (s_{out,1}, s_{out,2}))) \in Tra_{1||2}$$

- For all $e_1 \in E_1 \setminus E_2$ and $s_2 \in S_2$, if:
 $((s_{in,1}, guard_1, e_1, reset_1, s_{out,i}) \in Tra_1 \text{ then.}$

$$((s_{in,1}, s_2), guard_1, e_1, reset_1, (s_{out,1}, s_2)) \in Tra_{1||2}$$

- For all $e_1 \in E_2 \setminus E_1$ and $s_1 \in S_1$, if:
 $((s_{in,2}, guard_2, e_1, reset_2, s_{out,2}) \in Tra_2 \text{ then.}$

$$((s_1, s_{in,2}), guard_2, e_2, reset_2, (s_1, s_{out,2})) \in Tra_{1||2}$$

Hypotheses In this contribution, several hypotheses and restrictions are considered. They can be outlined as follows:

- Supply Chains must be reconfigurable: In a dynamic business environment, companies often need to modify their processes or partners to remain competitive. Reconfiguration facilitates the integration of new technologies or production methods.
- Systems must be able to select the appropriate scenario in case of disturbances: By incorporating systems capable of evaluating and selecting scenarios, companies can react swiftly to crises, which is essential in an uncertain economic environment.
- The SC behavior is determined by events, meaning that actions or changes in the system are triggered by specific events that occur at specific times. In summary, all models will be event-deterministic: By focusing on determining events, models become simpler and easier to understand, helping all involved to monitor and interpret system results.
- We will exclusively utilize a single clock for system operation. In other words, the treated SC will be managed by centralized approaches: A centralized approach facilitates resource management and decision-making, as it standardizes processes and avoids potential conflicts among different operational units.

3 Behavior and performance analysis of TAG

This section offers a formal modeling of TAG in dioid algebra, which was initially presented in Niguez et al. (2016). Furthermore, in order to improve efficiency and

respond quickly to customer demand, evaluating the performances of SC using the formalisms of dioid algebra is crucial.

When modeling a supply chain using TAGs formalisms, several key assumptions are made to effectively represent the dynamic and uncertain nature of real-world supply chains. These assumptions help capture the variability and uncertainty inherent in processes like transportation, production, and inventory management, while still providing a tractable mathematical framework for analysis and optimization. You will find below the assumptions taken into account for this study, as well as explanations of the reasons for their choice:

- i) **Discrete event system representation:** The supply chain is represented as a discrete event system where events (such as the arrival of goods, completion of production, or inventory updates) occur at specific moments in time. Supply chains consist of discrete processes (such as delivery, production, inventory management), which occur at specific points in time. Timed automata provide a natural framework for modeling such events, making it easier to track and control the sequence and timing of operations.
- ii) **Time intervals for transitions:** The transition times between different states (e.g., between processing steps, transport, or storage stages) are represented as time intervals, rather than fixed durations. In real-world supply chains, the time required to complete a task or move goods between stages often varies due to factors such as traffic conditions, production delays, or operational inefficiencies. Modeling transitions as time intervals allows the system to account for this variability, making the model more realistic and capable of capturing real-world uncertainties.
- iii) **Perturbations as external disturbances:** Perturbations, such as supply disruptions, transportation delays, resource shortages, or unexpected changes in demand, are modeled as external disturbances that affect the regular operation of the supply chain. Real-world supply chains are frequently affected by external factors (e.g., weather, geopolitical issues, or natural disasters) that cause unanticipated disruptions. By treating these disturbances as external events that affect the timing of processes, the model can simulate realistic supply chain behavior under stress or unforeseen conditions.
- iv) **Uncertainty in duration of perturbations:** The duration and impact of perturbations are uncertain and modeled as time intervals. For example, a disruption in transportation may take an unknown amount of time to resolve, and the delay could vary. Disturbances in real-world supply chains do not have fixed durations. For example, a port strike or weather-related delay may last a variable amount of time. Using time intervals or stochastic processes allows the model to handle the uncertainty in how long perturbations last and their potential impact on the system, which is crucial for decision-making under uncertainty.

3.1 Linear representation of TAG using dioid algebra

Definition 2 (Niguez et al. 2016). A $(\text{Max}, +)$ or $(\text{Min}, +)$ representation of TAG is described by a 7-tuple: $H = (S, E, Tra, \alpha, \beta, u_{sup}, u_{inf})$, where:

S : is the set of states;

E : is the set of events;

Tra : is the set of timed transitions of the automaton;

$\alpha \in D^{1 \times |Q|}$: is the initial delays such that if $s \in S_0$, then $\alpha_s = e$ else $\alpha_s = \varepsilon$;

$\beta \in D^{|Q| \times 1}$: is the final delays such that if $s \in S_f$, then $\beta_s = e$ else $\beta_s = \varepsilon$;

$$u_{sup} : E \rightarrow D^{|S| \times |S|} :$$

$$[u_{sup}(a)]_{s_{in}s_{out}} = \begin{cases} \sup(\text{guard}) & (s_{in}, \text{guard}, a, \text{reset}, s_{out}) \in Tra \\ \varepsilon & (s_{in}, \text{guard}, a, \text{reset}, s_{out}) \notin Tra \end{cases}$$

$$u_{inf} : E \rightarrow D^{|S| \times |S|} :$$

$$[u_{inf}(a)]_{s_{in}s_{out}} = \begin{cases} \inf(\text{guard}) & (s_{in}, \text{guard}, a, \text{reset}, s_{out}) \in Tra \\ \varepsilon & (s_{in}, \text{guard}, a, \text{reset}, s_{out}) \notin Tra \end{cases}$$

For each $(s_{in}, \text{guard}, e, \text{reset}, s_{out}) \in Tra$, the coefficients $[u_{sup}(a)]_{s_{in}s_{out}}$ and $[u_{inf}(a)]_{s_{in}s_{out}}$ correspond to the upper and lower bounds of guard intervals, respectively.

To represent the dynamic evolution of (Max, +) and (Min, +) automata (Gaubert 1995), two vectors $X_{sup}(\omega)$ and $X_{inf}(\omega)$ that are defined in $D^{1 \times |Q|}$ are introduced, with $\omega \in E^*$, $a \in E$ and $\alpha \in \mathbb{R}^{1 \times |Q|}$. They are also known as generalized dater limits, represented as the following:

$$\begin{cases} X_{sup}(\varepsilon) = \alpha \\ X_{sup}(\omega a) = X_{sup}(\omega) \otimes u_{sup}(a) \end{cases} \quad (1)$$

and

$$\begin{cases} X_{inf}(\varepsilon) = \alpha \\ X_{inf}(\omega a) = X_{inf}(\omega) \otimes u_{inf}(a) \end{cases} \quad (2)$$

In TAG, the maximum and minimum execution times of an event sequence are found using representations (1) and (2).

3.2 Representations evaluating the completion dates

In this section, several representations are involved with the aim of capturing the extreme behaviors of TAG. Noting that some studies have introduced a novel formalism in dioid algebra to assess the performances of systems modeled by TAG. They established recursive equations in (Max, +) and (Min, +) to depict the system behavior in both worst and best-case scenarios. By the process of induction, we give two dater vectors, $x_{sup}(n)$ and $x_{inf}(n)$, represented in $D^{|Tra| \times 1}$, where $|Tra| = m$ is the number

of TAG transitions (Ait-Oumeziane et al. 2020a):

$$\begin{cases} x_{sup(p,a,q)}(1) = \alpha_p & (p, a, q) \in Tra, \\ x_{sup}(n+1) = A_{sup} \otimes x_{sup}(n) & n \geq 1 \end{cases} \quad (3)$$

and

$$\begin{cases} x_{inf(p,a,q)}(1) = \alpha_p & (p, a, q) \in Tra, \\ x_{inf}(n+1) = A_{inf} \otimes x_{inf}(n) & n \geq 1 \end{cases} \quad (4)$$

The matrix $A_{sup} \in D^{m \times m}$ (resp. $A_{inf} \in D^{m \times m}$), with j and k respectively correspond to the timed transitions (p, a, q) and (r, a', s) in Tra , is designated as:

$$A_{jk,sup} = \begin{cases} [u_{sup}(a')](r,a',s) & s = p \\ \varepsilon & s \neq p, \end{cases} \quad (5)$$

and

$$A_{jk,min} = \begin{cases} [u_{min}(a')](r,a',s) & s = p \\ \varepsilon & s \neq p \end{cases} \quad (6)$$

The notation $A_{jk} = \varepsilon$ indicates that the transition (p, a, q) of index j in Tra can occur after the transition (r, a', s) of index k in Tra occurs.

It should be noted that D can stand for $\mathbb{R}_{max}$ when a TAG behaves in its worst way, and D defined as $\mathbb{R}_{min}$; when it exhibits its optimistic behavior.

After taking into account all of this, we can say that the minimum completion interval for $D=\mathbb{R}_{min}$ (resp. the maximum for $D=\mathbb{R}_{max}$) of all paths of length (n) that lead to state p is $[x_{inf}(n+1), x_{sup}(n+1)]$. Actually, the recurrences (3) and (4) in $\mathbb{R}_{max}$ are taken into consideration to determine the worst behavior of TAG, and in $\mathbb{R}_{min}$ to represent the optimistic behavior.

Proof (minimum completion interval). Applying recurrence, yields to:

For x_{sup} :

Base case

For any $p \in S$, we have by definition: $x_{sup}(1) = \alpha_p = X_{sup}(\varepsilon) = \{X_{sup}(\omega) \mid |\omega| = 0\}$.

Inductive case

For all sequence of length $n+1$ resulting in the state p , we have:

$\oplus_{\{\omega \mid |\omega|=n+1\}} X_{sup}(\omega) = \oplus_{q \in S} \oplus_{\{a \mid (q,a,p) \in Tra\}} \oplus_{\{\omega' \mid |\omega'|=n\}} X_{sup}(\omega') \otimes u_{sup}(a)_{qp}$
(according to Eq. (1)).

This results in the following

$$\begin{aligned} &= \oplus_{q \in S} (\oplus_{\{\omega' \mid |\omega'|=n\}} X_{sup}(\omega')) \otimes (\oplus_{\{a \mid (q,a,p) \in Tra\}} u_{sup}(a)_{qp}) \\ &= \oplus_{q \in S} x_{sup}(n+1) \otimes (\oplus_{\{a \mid (q,a,p) \in Tra\}} u_{sup}(a)_{qp}) \quad (\text{by inductive case}) \end{aligned}$$

$$\begin{aligned}
&= \oplus_{q \in S} \oplus_{a \in \{(q,a,p) \in Tra\}} u_{sup}(a)_{qp} \otimes x_{(q,a,p)_{sup}}(n+1) \\
&= \oplus_{(q,a,p) \in Tra} \oplus_{j \in Tra} A_{j,(q,a,p)_{sup}} \otimes x_{(q,a,p)_{sup}}(n+1)
\end{aligned}$$

(by Eq. 3)

$$= [A_{sup} \otimes x_{sup}(n+1)] = x_{sup}(n+2).$$

For x_{inf} : the definition of the data vector x_{inf} is identical to that of the vector x_{sup} using instead Eq. 4.

3.3 Modeling process and performance analysis for a supply chain

A structured step-by-step summary for both modeling and performance analysis is given in Fig. 4.

4 Conception and synthesis of the reconfiguration methods

In this section, the processes and approaches used to design and synthesize effective reconfiguration methods within supply chains are explored. Reconfiguration is

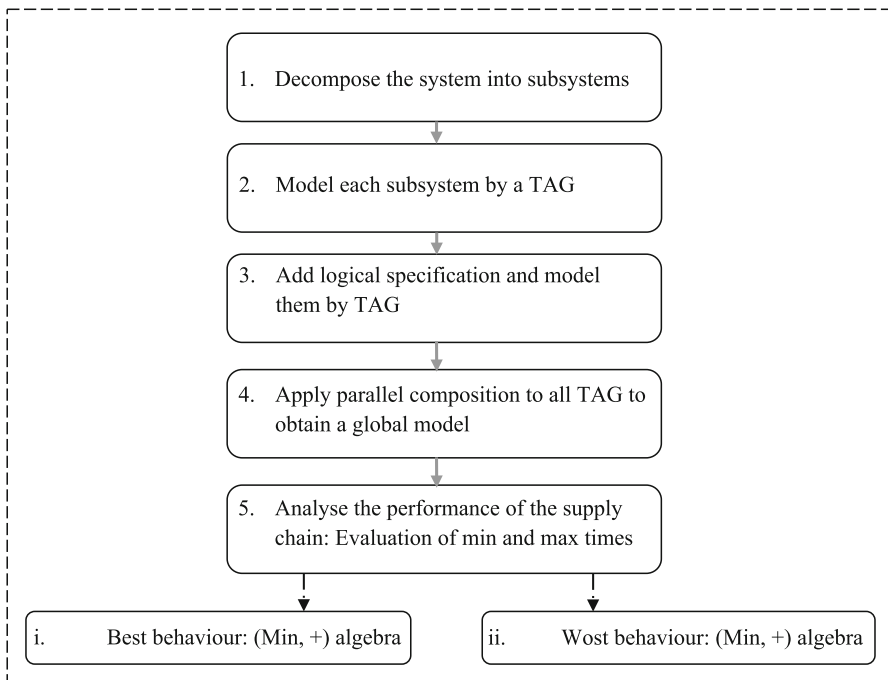


Fig. 4 Sequential overview of modeling and performance evaluation

essential to ensure that SC can adapt to changes, disruptions, and market evolution. First, the different methodologies available for carrying out this reconfiguration are examined. Next, the development of alternative scenarios, allowing for the consideration of various reconfiguration options, is discussed. Finally, the identification and implementation of the optimal solution among these scenarios are analyzed.

4.1 Supply chains reconfiguration methodologies

The aim of this section is to establish a methodology for constructing the reconfiguration model, denoted by $G(R)$, while representing the system behavior. To achieve this, two types of models are employed: Plant models and Specification models. These models are combined to generate the reconfiguration model.

The following step of the approach needs to be conducted offline.

4.1.1 Plant models

This is about representing the real functioning of the SC, including manufacturing processes, delivery times... This model takes into account the interactions between different actors in the SC, such as suppliers, distributors, etc., as well as external factors such as capacity constraints, and transportation delays. The TAG in Fig. 2 can be regarded as a plant model. It refers to the logical structure that describes how the plant SC interacts with its environment and evolves over time.

In the context of SC modeling and state transitions, events play a crucial role:

- Controllable events:
 - These events are immediately generated upon their expected occurrence.
 - They correspond to the current clock value $c(t)$ when entering a new state.
 - The interval associated with these transitions is designated by $c(t) = [c(t), c(t)]$.
- Uncontrollable events:
 - These events are anticipated to take place within a specified time frame.
 - They are expected to occur between 0 and μ .
 - The upper bound μ is unknown.

The interval in question represents the variability in the occurrence timing of these events.

4.1.2 Specification models

The specification models define the temporal requirements and expected behaviors of the SC from an abstract perspective. They are used to specify the intervals of transitions associated with uncontrollable events.

Figure 5 represents four specifications aimed at enforcing timed rules within the SC depicted in Fig. 1. $G(S_1)$ dictates that the event a should take place between 1 and 3 time units after the occurrence of event E_1 . Specification $G(S_2)$ ensures that event b occurs within 4 to 5 time units after event a . Furthermore, $G(S_3)$ specifies that the

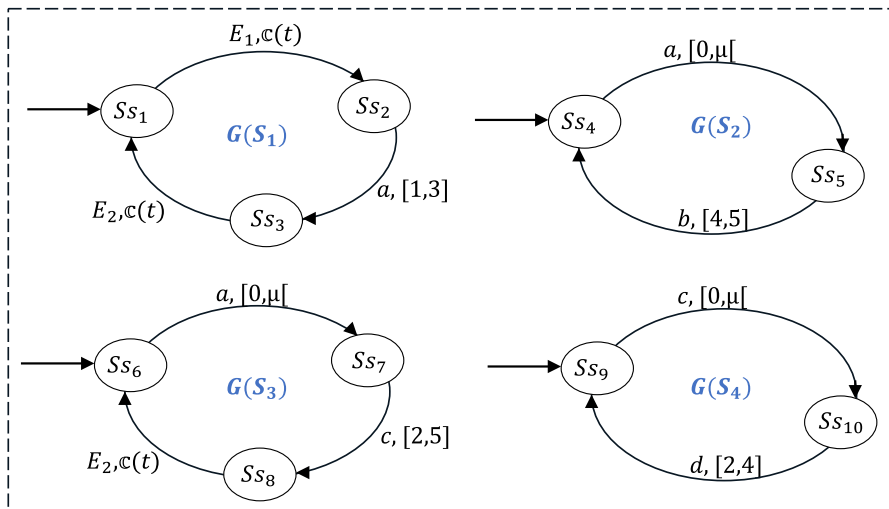


Fig. 5 Four timed specification models

event c must occur between 2 and 5 time units after the occurrence of event a . Lastly, $G(S_4)$ stipulates that event d must occur between 2 and 4 time units following the occurrence of event c .

4.2 Algorithmic instruction for the reconfiguration model

Reconfiguration models can become complex, especially in large systems or systems with many possible reconfiguration scenarios. Algorithms enable to manage this complexity by providing systematic and efficient methods for computing and representing the model. By using the plant and specification models explained in Sects. 4.1.1 and 4.1.2, the following algorithm is proposed to define the reconfiguration model:

Algorithm 1 Construction of the reconfiguration model of a supply chain

Given: plant models $G(P_i)$, specification models $G(Q_j)$
 Calculate the reconfiguration model $G(R)$ of the SC, defined by:

$$G(R) = G(P) \parallel G(Q)$$

 with $G(P) = \parallel_i G(P_i)$ and $G(Q) = \parallel_j G(Q_j)$
Result: reconfiguration model $G(R)$

4.3 Development of possible scenarios

By combining plant and specification models through parallel composition, we obtain a global reconfiguration model. This model represents the complete behavior of the system and describes all possible operating scenarios.

Given that the reconfiguration step is completed with the diagnosis results, it needs to be performed exclusively in online mode.

4.3.1 Analysis of non-occurring event: $\{e\}$

The following case deals with an event denoted by e , which, despite expectations, did not occur:

Algorithm 2 Reconfiguration process for the event $\{e\}$

- Given:** reconfiguration model $G(R)$, diagnosis after initial examination $\{e\}$
1. Remove transitions associated with the event $\{e\}$ that did not occur.
 2. If there exists a path from initial state to final state in the reconfiguration model:
 - i. Extract the path with the fastest execution time
 - ii. Represent the reconfigured Supply Chain, instructing how the system should be adjusted to address the absence of event $\{e\}$.
 3. If a final state cannot be reached:
 - i. Reconfiguration is not feasible due to lack of redundancy.
-

4.3.2 Analysis of late/early event: $\{e\}$

The following case, treats an expected event e , that happened too late or too early:

Algorithm 3 Reconfiguration process for the event $\{e, \text{early}\}$

- Given:** reconfiguration model $G(R)$, diagnosis residual $\{e, \text{late}\}$ or $\{e, \text{early}\}$
1. Modify the time limits of all transitions labelled with event e to ensure that the occurrence date of the event falls within the acceptable range.
 2. Find the fastest execution-time path from the initial state to the final state in the reconfiguration model.
- Results:** represent the reconfigured Supply Chain, ensuring the functioning of the SC
-

4.3.3 Unexpected event examination: $\{e\}$

Algorithm 4 illustrates a scenario in which an unexpected event e occurs:

Algorithm 4 Reconfiguration for the unexpected event $\{e\}$

- Given:** reconfiguration model $G(R)$, diagnosis of unexpected event $\{e\}$
 Implement a solution recommended by the model into the logistics chain.
- Results:** represent the reconfigured SC, while addressing the unexpected event
-

4.4 Optimal reconfiguration

For the optimal reconfiguration stage, two algorithms were developed. The first one selects all scenarios corresponding to the same disturbance $\{e\}$. This process will allow us to group and understand all the available alternatives in response to a specific disturbance. The second evaluates these scenarios. It analyzes each alternative based on

predefined criteria to determine the optimal options. By combining these two algorithms, all possible responses are not only identified to a disturbance but the most efficient solution is also chosen.

4.4.1 Exploration of the permissible reconfiguration set for $\{e\}$

We denote the set of scenarios containing all permissible transitions corresponding to a perturbation $\{e\}$, by S_e . The following algorithm describes the selection process of these reconfigurations:

Algorithm 5 Selection of all reconfigurations corresponding to the same event $\{e\}$

Given: reconfiguration model $G(R)$, diagnosis $\{e\}$

1. Initialize an empty set S_e to store scenarios correspond to $\{e\}$.
2. If s corresponds to $\{e\}$, for each scenario $s \in G(R)$:
 - i. Add s to the set S_e

Results: return the set S_e containing all scenarios that correspond to $\{e\}$

4.4.2 Evaluation of scenarios and choice of optimal reconfiguration

The optimal reconfiguration choice is based on nominal sequence execution time criteria using TAGs analytical representations with (Max, +) and (Min, +) algebra mathematical tools.

Algorithm 6 Optimal reconfiguration

Given: set of scenarios S_e

1. Go through each scenario $s \in S_e$.
2. Evaluate and calculate the nominal sequence execution time of each scenario.
3. Compare and keep track the scenario with the lowest nominal execution time encountered so far. This scenario corresponds to the optimal reconfiguration $G(Op)$.

Results: optimal scenario $G(Op)$

4.4.3 Reconfiguration procedure flowchart

In this section, we introduce a visual representation of the reconfiguration algorithms to aid in the understanding of their structure and operation. While the previous sections provide a thorough mathematical and theoretical explanation, the inclusion of a flowchart and pseudocode offers a more intuitive, step-by-step breakdown of the algorithmic processes. These visual aids are intended to clarify the sequence of operations, decision points, and transitions that occur during the execution of the algorithm.

The flowchart below presents an overview of the primary stages involved in the reconfiguration algorithm, illustrating how the system adapts and responds to various inputs and conditions. Each step in the flowchart corresponds to a specific action made by the algorithm, from initialization to the final reconfiguration outcome. The flowchart is designed to be an easily interpretable guide to the algorithm logic and flow (Fig. 6).

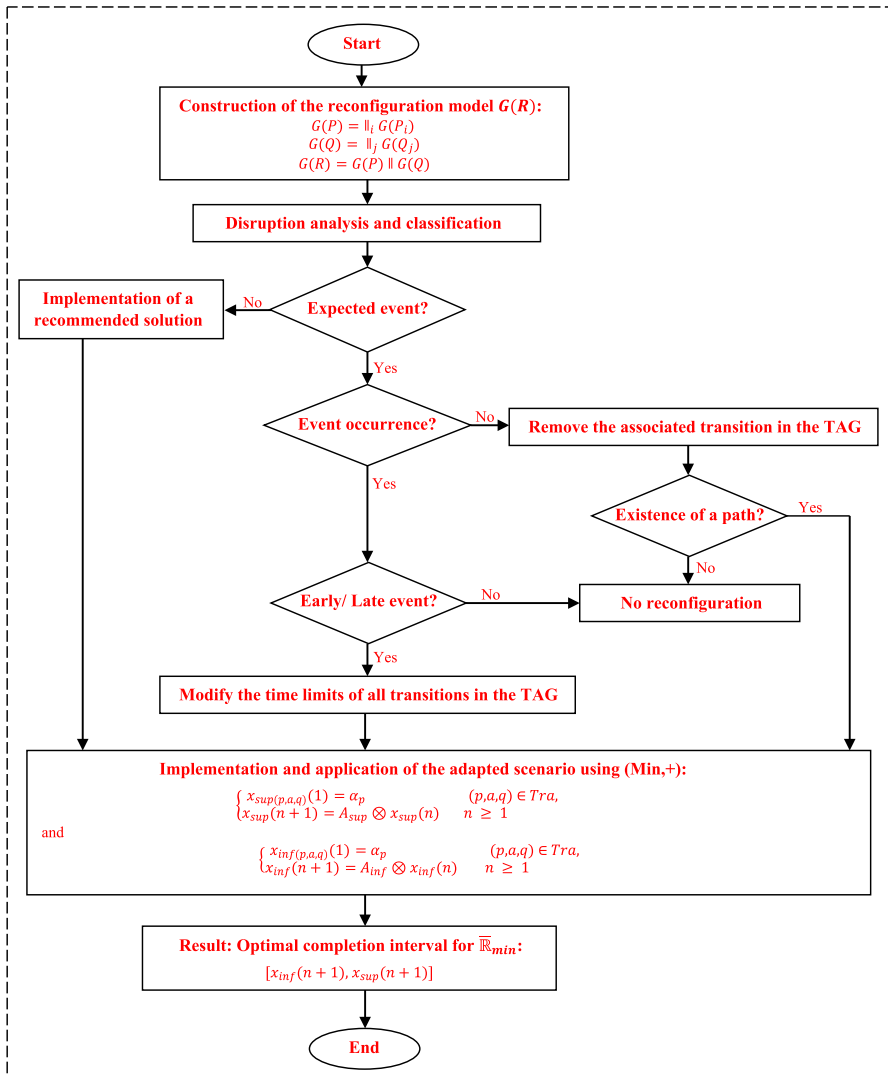


Fig. 6 Flowchart of the reconfiguration

4.5 Computational complexity analysis

The computational complexity of performance evaluation and reconfiguration methods for supply chains based on timed automata with guards and dioid algebra can be broken down into two primary parts: timed automata analysis and dioid algebra optimization. To provide a clear understanding, we will analyze each component in detail and how they interact.

The overall computational complexity depends on the combined impact of both components:

- The timed automata analysis step (reachability analysis or model checking) contributes exponential complexity in terms of the number of clocks and states, specifically $O(S \times 2^C)$, where S indicates the states and C is for the clocks.
- The optimization step using dioid algebra contributes polynomial complexity based on the supply chain graph size, specifically $O(n \times m)$ for single-source shortest path, $O(n^3)$ for all-pairs shortest path, or $O(n \times m^2)$ for max-flow problems, with n is the number of nodes (supply chain stages) and m is the number of edges (transitions between stages).

Final combined complexity:

- Worst-case complexity: The total complexity is dominated by the exponential factor from the timed automata analysis. Hence, the overall complexity is: $O(S \times 2^C + n \times m)$, for single-source shortest path; or $O(S \times 2^C + n^3)$ for all-pairs shortest path; or $O(S \times 2^C + n \times m^2)$ for max-flow.

5 Case study

In this section, the results of performance evaluation and reconfiguration are applied on a successful company specializing in delicious strawberry cakes “Mirabelle Enterprise” ((46) Le Développement Durable dans la PME de Mme MIRABELLE—YouTube). This company is defined as a SC, involving suppliers, operations and transportation, which fulfill their specific purpose to guarantee smooth and efficient logistics flows, while respecting the principles of the circular economy and environmental sustainability (see Fig. 7).

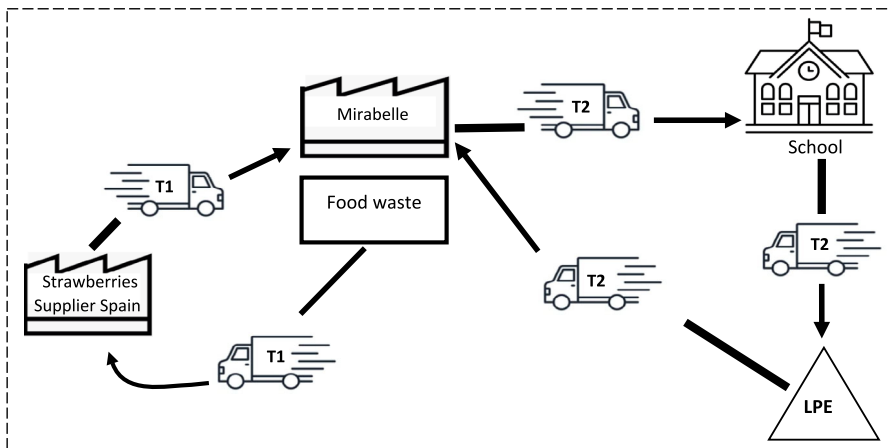


Fig. 7 The circular logistic ecosystem of Mirabelle Company

5.1 Description of the case study

To ensure a constant supply of quality strawberries, Mirabelle has established strong partnerships with a supplier in Spain “**Strawberries Supplier Spain**”. She utilizes an overseas carrier **T1** to deliver these strawberries to her business. Additionally, she composts the organic waste produced “**Food waste**” during production as part of her active commitment to environmental conservation. Its Spanish suppliers then recover this garbage to use in strawberry farming, promoting waste reduction and a circular economy.

In terms of internal logistics, Mirabelle has established a productive system for returning goods after the cakes are made. After delivering the cakes to the clients “**School**”, the truck **T2** goes back to a local producer to pick up the eggs “**LPE**”. By doing this, empty runs are avoided and the truck use is maximized.

5.2 The TAG supply chain model

A complicated system can be modeled using TAG by dividing it into multiple subsystems and modeling each one separately. The global model can be formed by integrating various subsystems through the use of parallel composition. For this, our SC will be decomposed into two fundamental components: the international logistic coordination and the streamlined internal logistic. Time intervals in TAG models are used to express uncertainty and denote the lower and upper boundaries of the event being studied.

The international logistic coordination: It is represented by the TAG in the Fig. 8, which is designated by “G1”:

S1: The strawberries are ready to be transferred by the Supplier from Spain.

S2: The storage state of strawberries in Mirabelle.

a: The transportation event of strawberries from Spain to Mirabelle.

b: The recuperation event of food waste by the suppliers of strawberries.

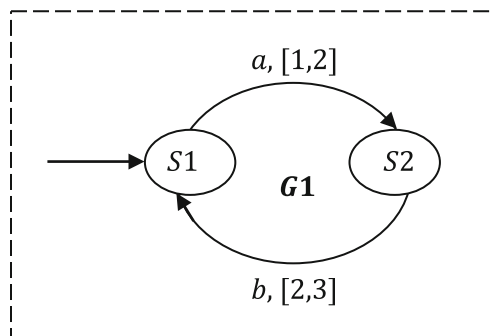
The streamlined internal logistic: The TAG “G2” in Fig. 9, depicted the movements between Mirabelle, the school and the local egg producer:

S3: Cakes are ready to be delivered from Mirabelle to the School.

S4: The presence of strawberry cakes in the School.

S5: The eggs are ready to be transferred to Mirabelle.

Fig. 8 The TAG of the international logistic coordination “G1”



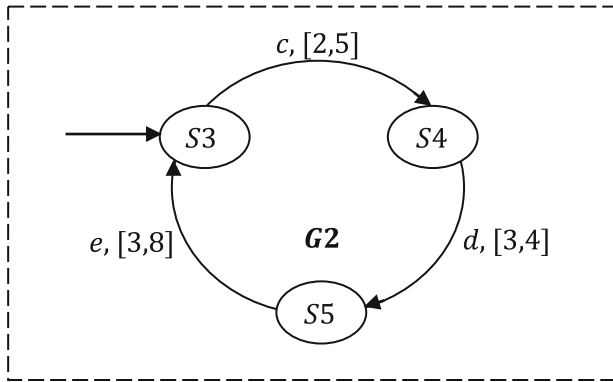


Fig. 9 The TAG of the streamlined internal logistic “G2”

c: The transportation event of cakes from Mirabelle to the school.

d: The journey between the school and the local egg producer (empty truck).

e: The recuperation event of eggs from the local producer to the company of Mirabelle.

The order specification: An additional specification automaton, identified as “G3”, must be included in order to preserve a logical flow within the SC (see Fig. 10). The TAG outline the sequential procedure that must be respected to prevent disruption and to ensure that the SC operates normally. It stipulates that the logistic must first go through the simultaneous presence of strawberries and eggs (event a and event e), followed by the transportation of food waste (event b) to Spain and the transportation of strawberries cakes to the School (event c).

The global model: We applied the parallel composition on the aforementioned TAG (G1, G2 and G3) to obtain the following TAG “G4” depicted in Fig. 11, which illustrates the various connections that could occur in the SC under study.

Actually, there are 48 transitions and 18 states in the SC global model. We note that the initial state, Q_0 , coincides with the final state.

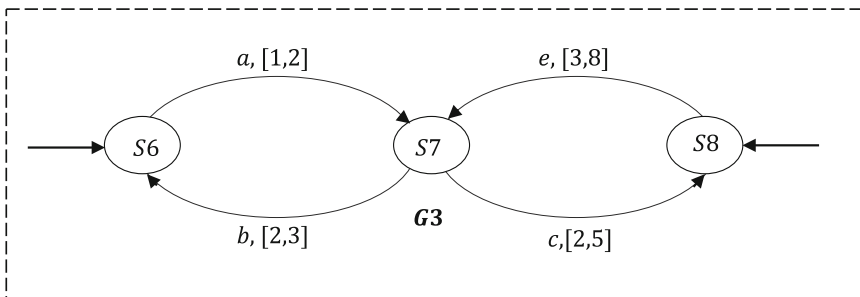


Fig. 10 The TAG of the main company “G3”

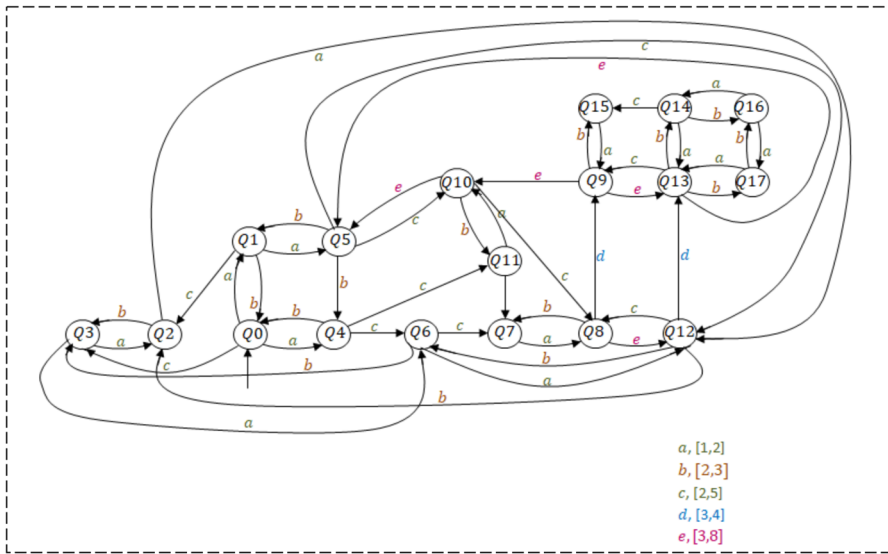


Fig. 11 The SC global model “G4”

5.3 Performances analysis of the modeled system

We identify the optimistic logistic chain completion intervals by using the above-mentioned representation of TAG with dioid algebra. The following is the set of transitions in “G4”, where $|Tra| = 48$:

$Tra = \{(Q0, a, Q1); (Q1, b, Q0); (Q1, c, Q2); (Q2, b, Q3); (Q3, a, Q2); (Q4, b, Q0); (Q0, a, Q4); (Q5, b, Q1); (Q1, a, Q5); (Q5, b, Q4); (Q4, c, Q6); (Q6, c, Q7); (Q6, b, Q3); (Q3, a, Q6); (Q0, c, Q3); (Q8, b, Q7); (Q7, a, Q8); (Q8, d, Q9); (Q9, e, Q10); (Q10, b, Q11); (Q11, a, Q10); (Q11, c, Q7); (Q10, c, Q8); (Q12, c, Q8); (Q8, e, Q12); (Q10, e, Q5); (Q5, c, Q10); (Q4, c, Q11); (Q5, c, Q12); (Q6, a, Q12); (Q12, b, Q6); (Q12, d, Q13); (Q2, a, Q12); (Q12, b, Q2); (Q13, b, Q14); (Q14, a, Q13); (Q14, b, Q16); (Q16, a, Q14); (Q17, b, Q16); (Q16, a, Q17); (Q14, c, Q15); (Q17, a, Q13); (Q13, b, Q17); (Q15, a, Q9); (Q9, b, Q15); (Q13, c, Q9); (Q9, e, Q13); (Q13, e, Q5)\}.$

A comprehensive manufacturing cycle must include all of the major logistic events in order to achieve the end state ($p = Q_0$) of the SC model. Since our model consists of 5 major events, our logistic cycle must have a length of 5 ($n = 5$) since our model consists of 5 major events.

We applied formulas (3) and (4) to find the generalized dater bounds $x_{sup}(n)$ and $x_{inf}(n)$ in $\mathbb{R}_{min}$ (Figure 13, Appendix). We get: $x_{inf}(6) = 7$ and $x_{sup}(6) = 12$.

In other words, for sequences of length 5 that lead to the final state Q_0 , the least interval of completion times is $[x_{inf}, x_{sup}] = [7, 12]$, i.e. the best path for the SC to achieve a production process is between 7 and 12 tu at the latest.

5.4 Reconfiguration of the treated SC

To enhance the understanding of the proposed method, we focus on reconfiguring a part of the relevant SC “G2”. This involves employing parallel composition techniques for three plant models (the transportation of cakes, the journey between the school and the producer of eggs and the transportation of eggs from the local producer to the company) and five specification models (delay of: cakes delivery, loading/unloading cakes, the journey between the school and the eggs producer, the eggs transportation and of loading/unloading eggs).

In this instance, activation delays of 2 time units affect the transportation of eggs from the local producer to the Company “e”. Applying the results given by Algorithm 4, transitions from Q_9 to Q_{10} , Q_9 to Q_{13} , Q_{13} to Q_5 and from Q_8 to Q_2 are all adjusted to “e”, [3, 10], causing a variation in the standard duration of the sequences that lead to the final state.

Consequently, we will select the path with the fastest execution time from the initial state to the final state within the reconfiguration model. As a result, the reconfigured SC satisfies Fig. 10 without states Q_2 , Q_3 , Q_4 , Q_6 , Q_7 , Q_8 , Q_9 , Q_{10} , Q_{11} , Q_{14} , Q_{15} , Q_{16} and Q_{17} .

As illustrated in Fig. 12, after the reconfiguration, certain parts of the supply chain (represented by the red crosses) have been deactivated or excluded from the model, altering the overall flow in the system. The green path shows the new optimal route through the reconfigured chain, minimizing time while avoiding the inaccessible or inefficient parts of the system.

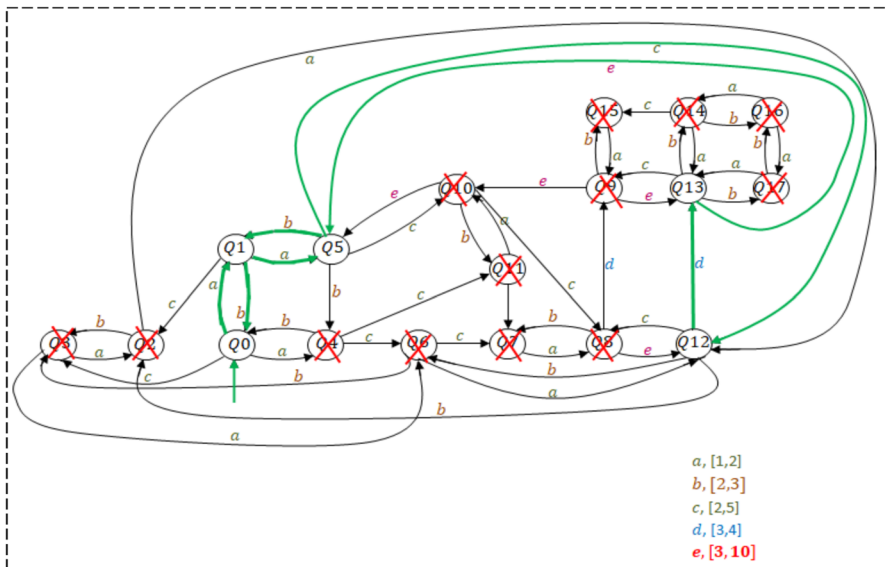


Fig. 12 Optimal path after supply chain reconfiguration

6 Conclusion

The objective of this article was to integrate two formal tools of DES to model and analyze the temporal performance of a SC. First, the SC was modeled by TAG. The method consisted, at modeling each subsystem as a separated TAG. Then, the global model was built using the parallel composition. Eventually, using dioid algebra, an analytical evaluation technique for identifying achievement dates inside the global TAG model is proposed.

The goal was to ensure that the production cycles were completed as soon as possible by optimizing the performance of the modeled SC. To deal with that, we employed the $(\text{Min}, +)$ algebraic formalisms of minimum completion intervals, which allowed us to determine the earliest completion dates. Our suggestion was to use $(\text{Max}, +)$ to describe maximum completion intervals in order to analyze the SC worst behavior concern.

Next, the issue of DES reconfiguration is the focus of this research topic. There are two steps involved in the reconfiguration procedure. The first step is the development of the reconfiguration model, representing the behavior of the entire system, respecting the specified rules and creating all possible scenarios. This is accomplished by the parallel composition of two types of models: the plant model and the specification model. The second step was based on the nominal time criteria of sequences execution and by using analytical representations of TAG, for selecting the optimal reconfiguration.

A practical example to highlight the frequency of uncertainties in delay decisions is carried out. Actually, the theoretical results are applicable when the system representation can be satisfied with a single clock. For this reason, future tracks could uncover more intricate distributed SC models, which may require extension to multiple-clock systems. Furthermore, computerizing the method into a tool format for use in industrial settings would be intriguing. Another important direction for future work is investigating the scalability of the proposed framework for large-scale supply chains. The distributed and modular nature of our approach, based on multi-clock timed automata with guards, offers several advantages, including improved scalability and manageability through modular design, reduced complexity by structuring the system into smaller components, and real-time adaptation to operational changes. Additionally, this approach enhances fault tolerance and resilience to disruptions, provides flexibility in reconfiguring supply chain components, improves decision-making and optimization at the module level, and ensures better coordination and synchronization across different entities. These characteristics make the framework particularly suitable for dynamic and complex supply chains. Future research could further explore its application to large-scale industrial supply chains and the development of automated tools to facilitate its practical implementation.

Appendix

See Fig. 13.

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Said Amari. Sid-Ali Addouche. Agnès Letouzey, Xavier Desforges. Kamal Medjaher. supervised the work, revised the manuscript, and provided constructive suggestions for improvement.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors declare that they have no competing interests.

Ethical approval This study did not involve human participants, animals, or any procedures requiring ethical approval.

Consent for publication Not applicable.

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